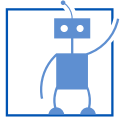


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Cognitive Systems: Neural Networks and Applied AI

Reinforcement Learning: Value Functions, Bellman Equations, Generalized Policy Iteration

Prof. Dr.-Ing. habil. Alois Knoll

Mohammadhossein Malmir, M.Sc.

Agenda

- **Value Functions**
- Bellman Equations
- Bellman Optimality Equations
- Generalized Policy Iteration (GPI)

Value Functions

- A reasonable strategy is to find policies π that maximizes the return G_t . For this purpose, we are interested in a **function that quantifies value for every state**.
- The **state-value function** is the expected return when a specific policy is followed after visiting a particular state:

$$v_{\pi}(s) \doteq \mathbb{E}_{\pi} [G_t | S_t = s]$$

- Similarly to the value function v , the **action-value function** q is the expected return when a specific policy is followed after choosing an action in a particular state:

$$q_{\pi}(s, a) \doteq \mathbb{E}_{\pi} [G_t | S_t = s, A_t = a]$$

*§Sutton, R. S. and Barto, A. G. Reinforcement learning: An introduction. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.58.
Slide is adapted from CogSys21 RL lecture held by Prof. Dr.-Ing. Noah Klarmann.*

π	policy	q	value for a specific action
A	stochastic action function	S	stochastic state function
a	particular action	s	particular state
G	return	v	value (expected return)

Agenda

- ✓ Value Functions
- **Bellman Equations**
- Bellman Optimality Equations
- Generalized Policy Iteration (GPI)

Bellman Equations

- An important property of the value function is the **recursive relationship** between consecutive states.
- Independently of the policy and the state, the following identity holds between the value of a state and the predecessor state (for all $s \in S$):

$$\begin{aligned}
 v_{\pi}(s) &\doteq \mathbb{E}_{\pi} [G_t | S_t = s] \\
 &= \mathbb{E}_{\pi} [R_{t+1} + \gamma G_{t+1} | S_t = s] \\
 &= \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) [r + \gamma \mathbb{E}_{\pi} [G_{t+1} | S_{t+1} = s']] \\
 &= \sum_a \pi(a|s) \sum_{s', r} p(s', r | s, a) [r + \gamma v_{\pi}(s')]
 \end{aligned}$$

π	policy
γ	discount rate
a	particular action
G	return
R	stochastic reward function
r	expected reward
s	particular state
S	stochastic state function
v	value (expected return)

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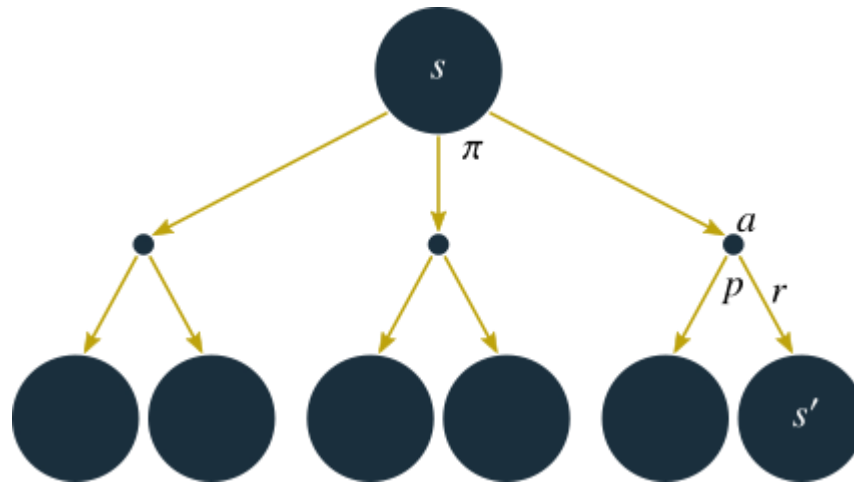
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Bellman Equations

- Bellman equation for state-value function:

$$v_{\pi}(s) = \sum_a \pi(a|s) \sum_{s', r} p(s', r|s, a) [r + \gamma v_{\pi}(s')]$$

- Backup diagram: shows relationships that form the basis of the updates for value-functions.



π	policy
γ	discount rate
s	particular state
a	particular action
r	expected reward
p	transition probability
q	value for a specific action
v	value (expected return)

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Bellman Equations

- Independently of the policy and the state, the following identity holds between the value of a state and the predecessor state (for all $s \in S$):

$$\begin{aligned}
 q_{\pi}(s, a) &\doteq \mathbb{E}_{\pi} [G_t | S_t = s, A_t = a] \\
 &= \mathbb{E}_{\pi} [R_{t+1} + \gamma G_{t+1} | S_t = s, A_t = a] \\
 &= \sum_{s', r} p(s', r | s, a) [r + \gamma \mathbb{E}_{\pi} [G_{t+1} | S_{t+1} = s']] \\
 &= \sum_{s', r} p(s', r | s, a) \left[r + \gamma \sum_{a'} \pi(a' | s') \mathbb{E}_{\pi} [G_{t+1} | S_{t+1} = s', A_{t+1} = a'] \right] \\
 &= \sum_{s', r} p(s', r | s, a) \left[r + \gamma \sum_{a'} \pi(a' | s') q_{\pi}(s', a') \right]
 \end{aligned}$$

π	policy
γ	discount rate
a	particular action
G	return
R	stochastic reward function
r	expected reward
s	particular state
S	stochastic state function
A	stochastic action function
q	value for a specific action

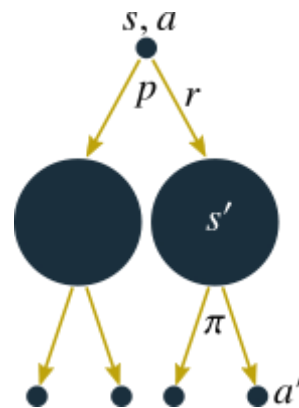
§Sutton, R. S. and Barto, A. G. Reinforcement learning: An introduction. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.61.

Bellman Equations

- Bellman equation for action-value function:

$$q_{\pi}(s, a) = \sum_{s', r} p(s', r | s, a) \left[r + \gamma \sum_{a'} \pi(a' | s') q_{\pi}(s', a') \right]$$

- Backup diagram: show relationships that form the basis of the updates for value-functions.



π	policy
γ	discount rate
s	particular state
a	particular action
r	expected reward
p	transition probability
q	value for a specific action
v	value (expected return)

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Agenda

- ✓ Value Functions
- ✓ Bellman Equations
- **Bellman Optimality Equations**
- Generalized Policy Iteration (GPI)

Bellman Optimality Equations

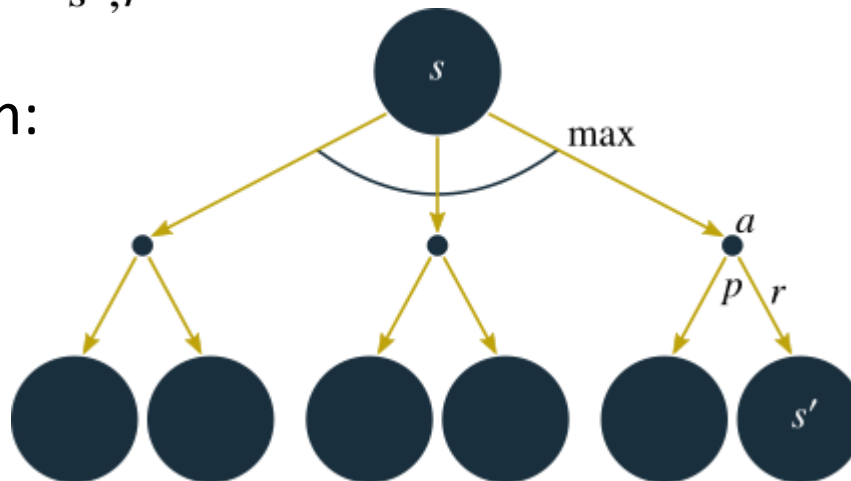
- Optimal state-value function: for all $s \in S$ $v_*(s) \doteq \max_{\pi} v_{\pi}(s)$

- Bellman optimality equation for state-value function:

$$v_*(s) = \sum_a \pi_*(a|s) \sum_{s',r} p(s',r|s,a) [r + \gamma v_*(s')]$$

$$= \max_a \sum_{s',r} p(s',r|s,a) [r + \gamma v_*(s')]$$

- Backup diagram:



π	policy
γ	discount rate
s	particular state
a	particular action
r	expected reward
p	transition probability
S	stochastic state function
q	value for a specific action
v	value (expected return)

§Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.62,63,64.

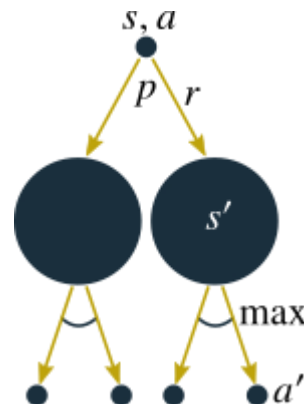
Bellman Optimality Equations

- Optimal action-value function: for all $s \in S$ $q_*(s, a) \doteq \max_{\pi} q_{\pi}(s, a)$
- Bellman optimality equation for action-value function:

$$q_*(s, a) = \sum_{s', r} p(s', r | s, a) \left[r + \gamma \sum_{a'} \pi_*(a' | s') q_*(s', a') \right]$$

$$= \sum_{s', r} p(s', r | s, a) \left[r + \gamma \max_{a'} q_*(s', a') \right]$$

- Backup diagram:



π	policy
γ	discount rate
s	particular state
a	particular action
r	expected reward
p	transition probability
S	stochastic state function
q	value for a specific action
v	value (expected return)



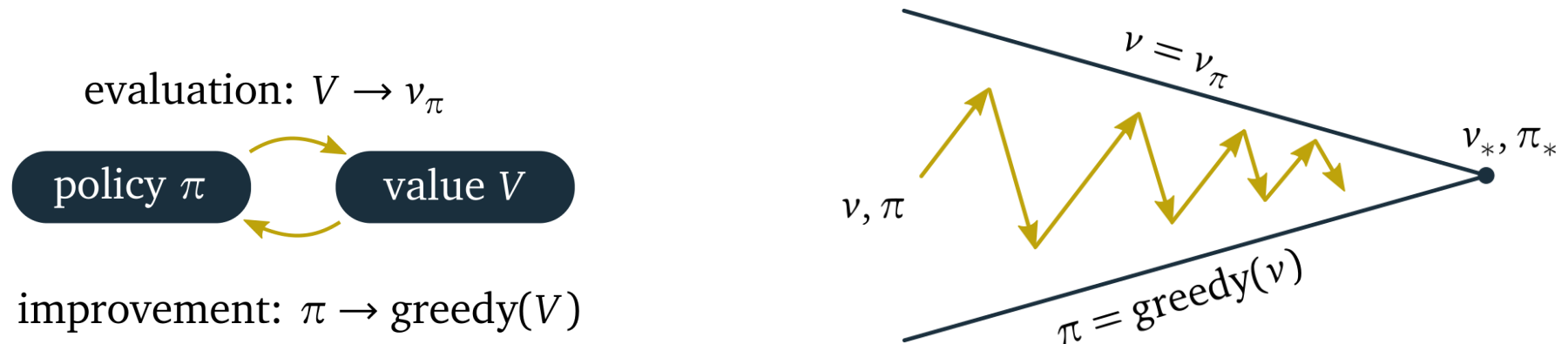
§Sutton, R. S. and Barto, A. G. *Reinforcement learning: An introduction*. 2nd ed. Adaptive computation and machine learning. Cambridge, Massachusetts: The MIT Press, 2018, p.63,64.

Agenda

- ✓ Value Functions
- ✓ Bellman Equations
- ✓ Bellman Optimality Equations
- **Generalized Policy Iteration (GPI)**

Generalized Policy Iteration (GPI)

- A value function depends on the policy; the policy depends on the value function. We hence need to iteratively apply value evaluation and policy improvement:



- Policy evaluation is called **prediction** (classical conditioning)
- Policy evaluation & improvement is denoted as **control** (operant conditioning)

π	policy
V	value estimates
v	value (expected return)

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